

LM04 Real Estate and Infrastructure

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1. Introduction

This learning module covers:

- Features and investment characteristics of real estate
- Features and investment characteristics of infrastructure

2. Real Estate Features

Real estate has two major sectors:

- **Residential:** Includes individual single-family detached homes and multi-family attached units owned by the residents. Residential real estate is the largest sector, making up some 75% of the market globally.
- **Commercial:** Commercial real estate primarily includes office buildings, shopping centers, and warehouses. When residential real estate properties (described above) are owned with the intent to rent, they are classified as commercial real estate.

Real Estate Investments

Exhibit 1 from the curriculum lists the main features of residential and commercial real estate.

	Residential real estate	Commercial real estate
Typical property	Owner-occupied, single residences; single-family residential property	Residential properties owned for lease or rental Office, retail, industrial, warehouse, hospitality, and mixed-use properties
Source of equity	Owners	Privately held by owners Publicly held through investors
Source of debt	Directly: Lenders (banks) through residential mortgages Indirectly: Investors in MBS that package residential mortgages	Directly: Lenders (banks) through commercial mortgages Indirectly: Investors in MBS that package commercial mortgages
Source of return to investors	Enjoyment of the property Price, or capital, appreciation	Income, or cash flow, generated by the property Price, or capital, appreciation

Real estate is different from other asset classes in several ways:

- Indivisibility – requires large capital investments
- Unique characteristics (no two properties are identical).
- There are various types of real estate investment alternatives available, ranging from

relatively liquid investments in stable, income-producing properties to illiquid investments over a long development life cycle spanning the purchase, construction/upgrade, occupancy, and sales phases.

- It can be difficult to diversify
- Indexes replicating the performance of real estate are not directly investable.

Also, the price discovery process in private real estate markets is opaque:

- Historical prices may not reflect current market conditions.
- Transactions are costly and time consuming.
- In some markets transaction activity may be limited due to supply or demand conditions.

Because of these unique features, real estate markets are typically fragmented. The local demand and supply conditions determine the value of a property, and local markets can be very different from national or global markets.

Real Estate Investment Structures

Real estate investing can be categorized along two dimensions: public/private markets and debt/equity based. Exhibit 2 presents the four quadrants for the basic forms of real estate investments with examples:

	Debt	Equity
Private	Mortgage debt	Direct ownership Sole ownership Joint ventures Limited partnerships
	Construction loans	Indirect ownership Real estate funds Private REITs
	Mezzanine debt	
Public	MBS/CMBS/CMOs	Publicly traded shares Construction Operating Development Public REITs
	Covered bonds	
	Mortgage REITs	
		UCITS/Mutual funds/ETFs
	Mortgage ETFs	

Equity-based investments represent ownership of real estate properties. Ownership can be through sole ownership, joint ventures, real estate limited partnerships, etc. A variation of equity-based investments is leveraged ownership: Assume a building costs \$10 million, and you put \$3 million of your money and borrow \$7 million. This is called leveraged ownership. That is, leveraged ownership is where a property is obtained through equity and mortgage

financing.

If you are investing in a **debt-based** real estate investment, it means you are lending money to a purchaser of real estate. A classic example is a mortgage loan. This is considered a real estate investment because the value of the mortgage loan is related to the value of the underlying property.

There can be many variations within the basic forms:

- Direct real estate investing: Involves purchasing a property and originating debt for one's own account. The major advantages are: control, and tax benefits. The major disadvantages are: extensive time and expertise required to manage the property, the large capital requirements, and highly concentrated portfolios.
- Indirect real estate investing: Pooled investment vehicles are used to access the underlying real estate assets. The vehicles can be public or private, such as limited partnerships, mutual funds, corporate shares, REITs, and ETFs.
- Mortgages: Represent passive investments in which the lender can expect to receive a predefined stream of payments over the life of the mortgage.
- Private fund investing styles: Most real estate private equity funds are structured as infinite-life open-end funds, which allow investors to contribute or redeem capital throughout the life of the fund.
- REITs: REITs combine the features of mutual funds and real estate. An REIT is a company that owns income-producing real estate assets. In REITs, average investors pool their capital to invest (take ownership) in several large-scale, diversified income-generating real estate properties. The REIT issues shares, where each share represents a percentage ownership in the underlying property. The income generated is paid as a dividend to the shareholders.

The main advantage of the REIT structure is that it avoids double corporate taxation. Normal corporations pay taxes on income, and then the dividend paid from the after-tax earnings are taxed again at the shareholder's personal tax rate. REITs can avoid corporate income taxes by distributing 90% - 100% of their rental income as dividends. Another advantage is that REITs are more transparent than private real estate markets.

The value of the REIT shares is based on the dividend. REIT shares often trade publicly on exchanges. It is a way for individual investors to earn a share of the income from commercial properties (office buildings, warehouses, and shopping malls) without buying them. Risk and return of REITs vary based on the types of properties they invest in.

Equity REITs invest in properties outright or through partnerships and joint ventures. The business strategy for equity REITs is simple: Maximize property occupancy rates

and rents while minimizing ongoing operating and maintenance expenses to maximize cash income and dividends. Mortgage REITs invest in real estate debt, typically MBS. Hybrid REITs invest in both real estate debt and equity.

3. Real Estate Investment Characteristics

The key reasons for investing in real estate are:

- Potential for competitive long-term returns (income and capital appreciation).
- Rent for long-term leases will lessen the impact of economic shocks.
- Diversification because of low correlation with other asset classes such as stocks, bonds.
- Inflation hedge.

Source of Returns

The return on real estate investments comes from income or asset appreciation or a combination of both. More than half the returns earned by commercial real estate investors comes from income, and it is a more consistent source of return throughout an economic cycle, as compared to capital appreciation.

Real estate investments can be similar to bond investments in that they generate stable, predictable, lower-risk cash flows from leases that are comparable to bond coupon payments.

Real estate investments are similar to equity investments in that they provide speculative returns based on the price appreciation of the real estate asset.

Real Estate Investment Diversification Benefits

Many investors prefer real estate for its ability to provide high, steady current income. Real estate also has low correlation with other asset classes and thus provides diversification benefits. However, there are periods when equity REIT correlations with other securities are high, and their correlations are highest during steep market downturns.

4. Infrastructure Investment Features

The assets underlying infrastructure investments are real, capital intensive, and long-lived. These assets are intended for public use, and they provide essential services e.g., airports, health care facilities, and power plants.

Infrastructure Investments

Like real estate, investment in infrastructure involves acquiring unique, illiquid assets with distinct locations, features, and uses. The returns can come from income and capital appreciation.

Rather than rentals, infrastructure cash flows arise from contractual payments such as:

- Availability payments: Payments received to make the facility available.

- Usage-based payments: Tolls and fees for using the facility.
- “Take-or-pay” arrangements: This obligates buyers to pay a minimum purchase price to sellers for a pre-agreed volume.

Infrastructure assets were primarily owned, financed, and operated by the government. Of late, they are financed privately through the use of public-private partnerships (PPPs). A public-private partnership is typically defined as a long-term contractual relationship between the public and private sectors for the purpose of having the private sector deliver a project or service traditionally provided by the public sector.

Categories of Infrastructure Investments

Infrastructure investments may be categorized based on: (1) underlying assets, and (2) stage of development of the underlying assets.

Infrastructure investments based on underlying assets: They can be classified into economic and social infrastructure assets.

- Economic infrastructure assets: These include transportation, communication, and social utility assets that are needed to support economic activity. Examples of transportation assets are roads, airports, bridges, tunnels, ports, etc. Examples of utility assets are assets used to transmit and distribute gas, electricity, generate power, etc. Examples of communication assets are assets that are used to broadcast information.
- Social infrastructure assets: These are assets required for the benefit of the society such as educational and healthcare facilities.

Infrastructure investments based on the stage of development of the underlying assets: They can be classified into greenfield, brownfield and secondary-stage investments.

- Greenfield investments develop new assets and new infrastructure. The objective may be to construct and sell the assets to the government, or to hold and operate the assets.
- Brownfield investments expand existing facilities and may involve privatization of public assets. They have a shorter investment period and immediate cash flows. The assets may also have a financial and operating history.
- Secondary-stage investments invest in existing infrastructure facilities or fully operational assets that do not need further expansion and investments. They generate immediate cash flows.

Forms of Infrastructure Investments

Investors may invest either directly or indirectly in infrastructure investments. The investment form affects the liquidity and the income and cash flows to the investor.

The advantages of investing directly in infrastructure are that investors have a control over

the asset and can capture the full value of the asset. But the downside of a large investment is that it results in concentration and liquidity risks.

Most investors invest indirectly. Some forms of indirect investments include:

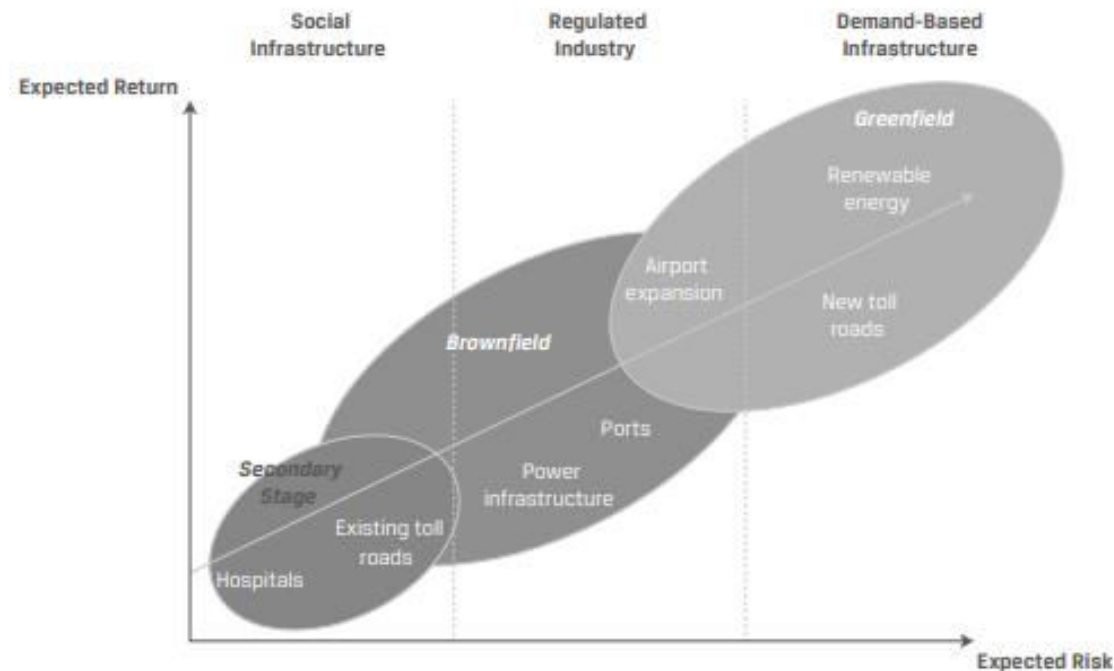
- investment in an infrastructure fund
- infrastructure ETFs
- shares of companies

Investing in publicly traded infrastructure companies offer the benefit of liquidity. Publicly traded infrastructure securities also have a reasonable fee structure, transparent governance, and provide the benefit of diversification. Master limited partnerships (MLPs) are pass-through entities similar to REITs and are listed on exchanges.

Infrastructure debt financing can take the form of both private and publicly traded debt. The terms are typically flexible to accommodate periods of zero cash flow and long development or investment horizons. Because of the stable underlying nature of cash flows, infrastructure debt has lower default rates and higher recovery rates than comparable fixed-income instruments, and it is less volatile over the economic cycle.

5. Infrastructure Investment Characteristics

Exhibit 8 from the curriculum depicts the investment characteristics of different types of infrastructure investments.



Greenfield infrastructure investments have the highest expected return and the highest expected risk, while secondary stage investments have the lowest expected return and the

lowest expected risk.

Also, investments in basic social services infrastructure (e.g., hospitals) or existing regulated industries (e.g., Power grids) typically involves lower risk and lower expected return. Whereas, demand-based infrastructure (e.g., new toll roads) are built on projections of future economic growth and are riskier.

Infrastructure Diversification Benefits

Some of the advantages to investors from investing in infrastructure are as follows:

- a steady income stream
- potential for capital appreciation
- diversification because of low correlation of infrastructure assets to traditional investments
- protection against inflation
- match the long-term liability structure of some investors such as pension funds

Summary

LO: Explain features and characteristics of real estate.

Real estate includes two major sectors: residential and commercial.

Unique features of real estate are: heterogeneity, fragmentation, challenges in price discovery, costly and time-consuming transactions.

LO: Explain the investment characteristics of real estate investments.

Real estate investing can be categorized along two dimensions: public/private markets and debt/equity based.

The return on real estate investments comes from income or asset appreciation or a combination of both.

Real estate investments can be similar to bond investments in that they generate stable, predictable, lower-risk cash flows from leases that are comparable to bond coupon payments. Real estate investments are similar to equity investments in that they provide speculative returns based on the price appreciation of the real estate asset.

Real estate provides diversification benefits because of their low correlation with other asset classes. However, during steep market downturns, equity REIT correlations with market benchmarks can increase.

LO: Explain features and characteristics of infrastructure.

The assets underlying infrastructure investments are real, capital intensive, and long-lived. These assets are intended for public use, and they provide essential services.

Infrastructure investments can be categorized based on:

- underlying assets as either economic or social
- stage of development of underlying assets as greenfield, brownfield, or secondary stage.

Infrastructure assets were primarily owned, financed, and operated by the government. Of late, they are financed privately through the use of public-private partnerships (PPPs).

LO: Explain the investment characteristics of infrastructure investments.

Greenfield infrastructure investments have the highest expected return and the highest expected risk, while secondary stage investments have the lowest expected return and the lowest expected risk.

Some of the advantages to investors from investing in infrastructure are as follows:

- a steady income stream
- potential for capital appreciation
- diversification because of low correlation of infrastructure assets to traditional

investments

- protection against inflation
- match the long-term liability structure of some investors such as pension funds